THE LAST MILE

The last mile of a diabetes result.

The lab calls the doctor. Nobody calls the patient.

A closure ledger for abnormal and critical diabetes results. A named owner, a stage clock, a read-back, an exception, and a receipt.

IIT Bombay Health-a-thon 2026
Diabetes track · Build Sprint

The lab is done when it delivers. The clinic is done when the patient is reached.

CLOSURE RECEIPT

Asha Rao · P0482

Potassium 5.9 mmol/L · ACE inhibitor start

03:10	**Lab releases the report**	Critical, lab flagged
03:12	**Clinician sees one line**	One tap, no form
03:40	**Nurse records the read-back**	Call · caregiver confirmed
03:55	**Register prints the closure**	Owner, clock, receipt

CLOSURE CERTIFICATE

Chain intact · evidence present · verify the live receipt

Unreachable is a disposition, not a false informed.

THE RESULT STOPS AT THE DOCTOR

Only one in four critical values reached the treating clinician.

25.8%

critical values reached a clinician

52.9%

went to nursing staff

90%

of never-notified values were outpatients

A 2015 audit of 1,187 critical values at an Indian tertiary hospital found a written policy, a log book and a read-back rule. The failure was not the rule. It was the last mile: the result reached a desk, not the person who could act. The same paper found 9 in 10 never-notified values were outpatients.

LAB
releases the report

→

DOCTOR
echoes the value

→

PATIENT
is contacted, or not

14% of positive results never returned (NMJI 2015). 6.8% to 62% of abnormal ambulatory results were not followed up (J Gen Intern Med 2012).

CLOSE THE LAST MILE, NOT ANOTHER INBOX

Four surfaces. One obligation.

The doctor sees one line. The nurse records the read-back. The desk resolves identity. The quality head prints the register.

01 CLINICIAN LINE

One card, one tap. Seen, corrected with a reason, or refused. No patient-side form.

02 NURSE CALLS

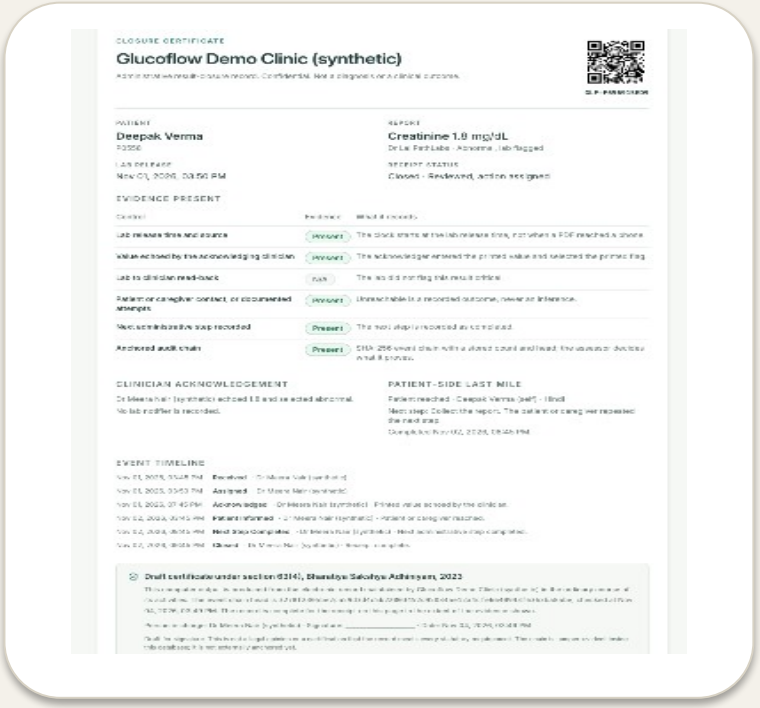
Read-back, teach-back, outcome, next step. An informed outcome needs the read-back.

03 DESK INTAKE

Accept custody, match a candidate with evidence, assign. An unmatched report cannot be assigned.

04 QUALITY REGISTER

A4 print, exceptions, Close. The page the surveyor files.


CLOSURE CERTIFICATE

Glucoflow Demo Clinic (synthetic)

PATIENT
Deepak Verma

REPORT
Creatinine 1.8 mg/dL

EVIDENCE PRESENT

CLINICIAN ACKNOWLEDGEMENT

PATIENT-SIDE LAST MILE

EVENT TIMELINE

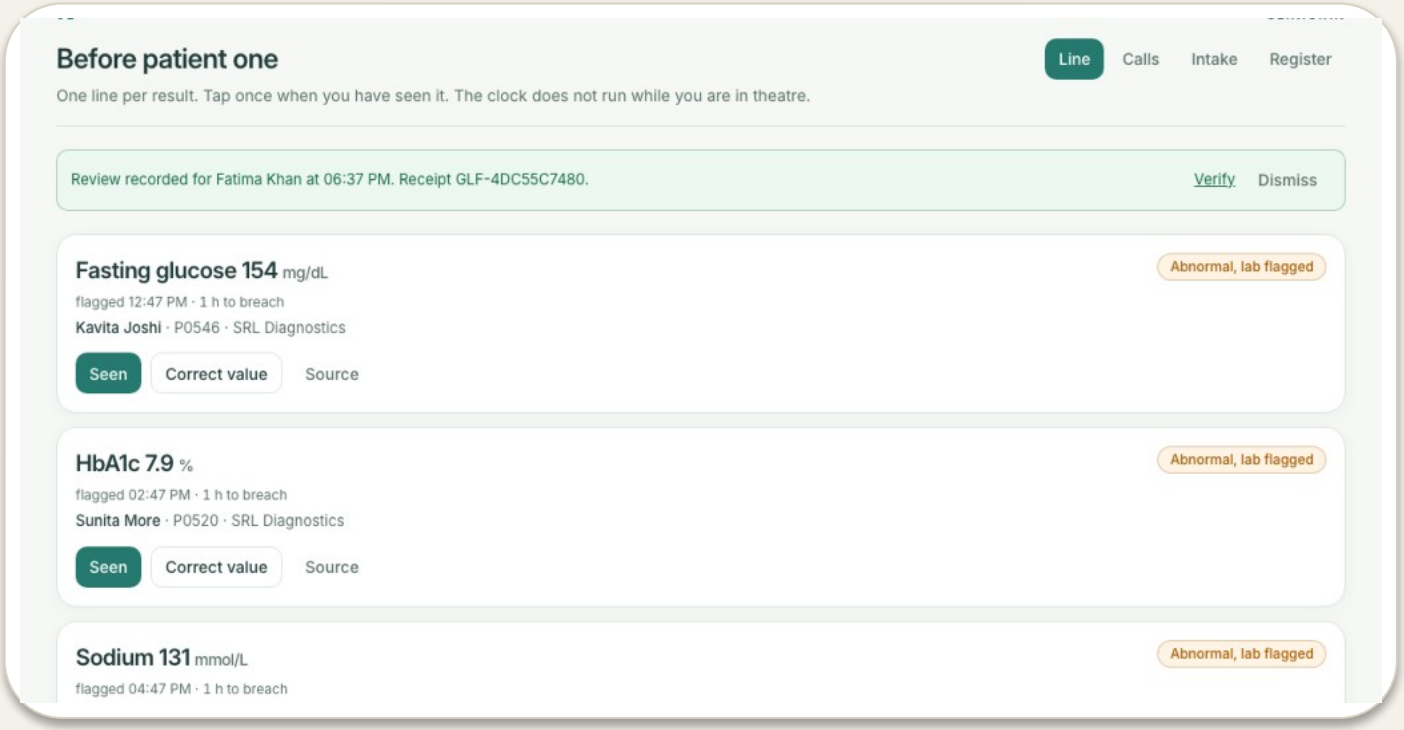
Draft certificate under section 63(4), Bharatiya Sakshya Adhiniyam, 2023

Closure certificate: evidence, timeline, section 63(4) draft.

THE LOOP RUNS TODAY

A working closure workspace.

Twenty synthetic diabetes receipts. Four role homes. One state machine. The line is one tap; the register prints; the replay is labelled synthetic.



ONE-TAP LINE

4 standing cards

READ-BACK

informed needs it

IDENTITY HOLD

match before assign

REGISTER

A4, exceptions, close

REPLAY

modelled, labelled

RECEIPT

chain + public verify

No live LIS or WhatsApp. The production adapter is the next gate.

THE RESULT A DIABETES DRUG MAKES NECESSARY

Start with the potassium and eGFR after a new ACE inhibitor.

The clinician sets the protocol. The lab flag sets criticality. The product does not infer a screening deadline or a clinical action.

01

START OR TITRATE

ACE inhibitor or ARB

The prescriber sets the monitoring protocol.

02

2 TO 4 WEEKS

Potassium + eGFR

The due window is a protocol, not an inference.

03

RESULT ARRIVES

Lab flag copied

Critical, abnormal or normal. No interpretation.

04

CLOSE THE LOOP

One tap + read-back

Patient reached, or attempts documented.

05

NEXT STEP

Repeat, review, attend

Tracked to completion; certificate ready.

The same receipt covers SGLT2i, MRA, annual uACR/eGFR, a critical potassium, or an abnormal HbA1c. 101 million people in India live with diabetes; the first market is the lab that already sends the report.

MEASURED, AND WHAT IS NOT

The build is measured. The demand is not.

WHAT WE RAN

212	unit and integration tests, 33 files
19	browser checks across four specs
0	lint findings on ledger, drawer, certificate, slip and verifier
12 / 12	database isolation checks
20	synthetic receipts, 10 patients, 4 labs

WHAT WE HAVE NOT VALIDATED

— No buyer interview, baseline audit, paid pilot or LOI.

— No live LIS, HL7/FHIR or WhatsApp send.

— No consent capture, DPA or retention runbook.

— No external anchor: the chain is tamper-evident inside one database.

— No clinical validation. No diagnosis, triage or treatment advice.

We will not claim these.

The next number is a baseline audit, not a TAM.

SELL THE CLOSURE RATE, NOT ANOTHER SEAT

One lab chain. Many clinics. A measurable last mile.

THE BUYER

Lab chain quality head, or a hospital group quality cell.

The lab already sends the report. It pays for proof that the last mile closed and for a differentiator to referring doctors. The clinic uses the inbox free; the diabetes clinician is the champion.

PILOT

₹60,000 / 90 days. One lab, one diabetes OPD, one ward. 100 de-identified reports for the baseline.

THE METRIC

Closed within the stage clock.

1. Abnormal or critical result with a named owner
2. Read-back recorded before an informed outcome
3. Patient reached, or attempts documented
4. Register printed; exception visible

The network is the expansion, not the product.

No TAM claim. The first proof is one lab, one cohort, one signed continuation.

THREE SITES. ONE MEASURED CLOSURE RATE

A baseline audit, not a demo.

100 consecutive abnormal or critical diabetes reports per site. Eight weeks. A baseline audit, then the closure workflow, then a paid continuation. If the first two gates fail, the thesis stops.

01 **ONE LAB OR OPD SPONSOR**
A quality head who will share 100 de-identified reports.

02 **ONE CLINICAL LEAD**
A practising diabetes clinician who signs the protocol and the pilot.

03 **ONE HONEST NUMBER**
Closed within SLA, exceptions documented, zero false informed.

THE CLOSING LINE

The lab is done when it delivers. The clinic is done when the patient is reached.

The result nobody told.

Synthetic demo. No real patient data. No diagnosis or treatment advice.